Test Normal Cases

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Test Normal Cases

> Prepare Test Normal Cases: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Plan routine cases with **expected result**s derived from the requirements

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Build a Paper Prototype

Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

Look for: Build a runnable paper prototype with explicit instructions and visible state

Reference example, not the only acceptable wording: The kit contains labelled instructions and catalogue, request fields, a marker and a trace sheet with run and version labels.

The request and catalogue stay fixed during the run; the current row and trace advance. The final **output** starts clear.

The cards validate, check both conditions, return the first fit, advance after failure and stop with no match after exhaustion.

Choose Inputs and Outputs

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

Look for: Turn the project outcome into precise **input** and **output** requirements

Reference example, not the only acceptable wording: Time required must be no more than available time; when quiet is required the activity must be quiet. Choose the first compatible activity in B1 to B4 order.

No means quiet is not required, not forbidden, so quiet B1 is permitted if 10 minutes fits.

Verify Important Outputs

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

Look for: Separate an answer into claims and connect each claim to relevant **evidence**

Reference example, not the only acceptable wording: A directly stated fact is four rolls, 2 m per roll or twelve 50 cm pieces.

A derived quantity is 8 m starting length, 6 m used or 2 m remaining, supported by the relevant calculation from R. R contains no enjoyment **evidence**, so citing it does not support everyone enjoying the activity.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

Concept to explain

Repeated success on ordinary cases is necessary but not sufficient.

Term	Meaning
test case	planned input and expected behavior
expected result	prediction written before running
actual result	observed behavior
reproducible	another tester can repeat the setup

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

The prototype remains an explicit rule system. This lesson focuses on prewritten expectations, repeatable conditions and honest records. Q version 0 and its comparison results are fictional worked examples.

Pupils must record their own P1 **observation**s, including operator errors, without copying invented successes. Judge both the **output** and its reason.

Routine **coverage** is deliberately limited; boundary and variation tests follow next week.

Readiness answer

The **expected result** comes from the requirements; the observed result records what the saved version actually did. Disagreement is **evidence** to investigate, so neither record should be silently changed to match the other.

The saved P1 version 1 cards, catalogue B version 1 and manual checklist from Week 32

Paper test sheets with case ID, **input**, purpose, **expected result**, observed result, judgement and trace reference

A folder for all versions and results; keep any reserved final evaluation cases separate from development work

End goal for the pictorial case

By the end, I can test routine cases against expectations and count full passes honestly.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Test Normal Cases using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT planned test **input** and **expected result**

2. Process — the deciding step

PROCESS run unchanged prototype and observe

3. Output — the result

OUTPUT actual result, pass/fail, and notes

Key idea

CORE IDEA Repeated success on ordinary cases is necessary but not sufficient.

How the mechanism works

UNDER THE HOOD A **test case** includes **input**, setup, expected behavior, actual behavior, and result. Precommitting expectations reduces hindsight bias.

Repeated normal tests establish basic reliability but say little about uncommon conditions.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE The sorter is tested on five common classroom items with expected bins written before each run.

Step 1

STEP 1 — OBSERVE planned test **input** and **expected result**

Step 2

STEP 2 — TRACE run unchanged prototype and observe

Step 3

STEP 3 — CONCLUDE **actual result**, pass/fail, and notes

Model answer

MODEL REASONING Write the expected **output** first, keep the prototype version fixed, record every actual **output**, and count passes without deleting failures.

Explain that five normal successes are necessary **evidence**, not full proof.

A claim to avoid

DO NOT OVERCLAIM Do not change the prototype halfway through a **test set** and combine results as if one version was tested.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

Four routine cases are chosen before testing. Three are correct.

What can be reported?

Exact explanation to model

Why the result follows

3/4 on those cases. Preserve the failure and trace its cause. This does not guarantee unfamiliar-case performance.

Explain the added worked example

Explain the added worked example · 1

This week you will turn separate paper traces into a test record. Choose several ordinary requests, write the required answers first, then run the same saved prototype on every request.

Testing is useful when it can reveal a mistake, so keep failures as carefully as successes. A demonstration chosen because it works is not a complete evaluation.

Explain the added worked example · 2

Use catalogue B version 1 in order: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These fictional values assume materials are available. **Input**s are whole minutes from 0 to 30 and Quiet required Yes or No. Time must fit AND either quiet is not required or the activity is quiet. No allows either kind.

Select the first fit with a correct reason; otherwise report no match. Missing or invalid **input** requires clarification. No booking or publishing takes place.

Explain the added worked example · 3

Keep the P1 version 1 instructions from Week 32: validate and stop with clarification if **input** is invalid; otherwise start at B1, check both conditions, suggest the first passing row and stop.

After a failure advance one row; report no match only after all rows fail. Reset to validation and B1 for each new valid request. The manual **baseline** marks every row against both conditions and selects the first fit.

Both methods must follow the same contract.

Explain the added worked example · 4

Give each case an ID. Record its exact **input**, purpose, expected **output** and expected explanation before running it. Then add observed **output**, observed explanation, pass or fail, and a trace reference.

Put the prototype version, catalogue version and **baseline** method at the top of the sheet.

A pass needs the correct result and a reason supported by the checked conditions; the right activity with a false reason is not a full pass.

Explain the added worked example · 5

Our routine set R contains four valid requests. R1 is 11 minutes Yes, checking an early quiet match. R2 is 9 minutes No, checking that the search reaches a later **fitting** row.

R3 is 6 minutes No, checking that a valid request can exhaust the list. R4 is 9 minutes Yes, checking that a time fit alone is insufficient. These cases do not cover every accepted **input** or validation error.

Key terms

Term	Meaning
Test case	A specified input with an expected result and a reason for testing it
Test set	A collection of cases used together to check a version
Pass criterion	The requirement an observed result must meet to count as a pass
Coverage	Which conditions or behaviours the selected tests examine
Baseline	A simple comparison method tested under the same conditions

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Case and request	Expected output and reason	Fictional observed Q version 0
R1 11 minutes Yes	B1; 10 fits 11 and B1 is quiet	B1 with the same correct reason; pass
R2 9 minutes No	B2; 8 fits 9, quiet unrestricted, B1 too long	B2 with the same correct reason; pass
R3 6 minutes No	No match; all activities take longer	No match with the same correct reason; pass
R4 9 minutes Yes	No match; B2 is not quiet and other rows take longer	B2 because 8 fits 9; fail

Model the reasoning aloud · 6

This is an invented record for a comparison prototype Q version 0, not a claim about your P1 runs. Q ignores the quiet condition when searching. Its trace explains the R4 failure.

In the same fictional comparison, the manual **baseline** produced the **expected result** and correct explanation on all four cases: 4 of 4 full passes. Q achieved 3 of 4. The denominator includes the failed case.

Model the reasoning aloud · 7

On this set Q has fewer full passes than the **baseline**. This supports investigating its quiet check; it does not establish performance on all requests. Do not claim a speed advantage because we did not measure time.

Correct implementations of P1 and the **baseline** can both pass all four, in which case these results show no correctness advantage for either method.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Test Normal Cases

A fictional pencil-pouch finder has been run on four ordinary requests.

The facts we will use

> **Supplied facts**
>
> Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9.

Supplied facts

Requests 2,4,6,8; recorded **output**s P1,P2,P1,none. These are invented records, not class results.

AI ACADEMY / AI-1/33

Expected versus recorded routine outputs

	Request	Expected	Recorded	Result
Test 1	2	P1	P1	pass
Test 2	4	P2	P2	pass
Test 3	6	P2	P1	fail
Test 4	8	none	none	pass

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/33

Test Normal Cases

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Write expectations

For 2,4,6,8 expect P1,P2,P2,none.



2. Compare outputs

Runs at 2,4 and 8 agree; run at 6 disagrees.



3. Trace failure

P1 holds only 3, so it cannot satisfy request 6.



4. Report

Three of four routine cases pass; retain the failure.

CHANGE ONE THING

Prototype reports none for request 6 instead of P1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Write expectations	For 2,4,6,8 expect P1,P2,P2,none.
2. Compare outputs	Runs at 2,4 and 8 agree; run at 6 disagrees.
3. Trace failure	P1 holds only 3, so it cannot satisfy request 6.
4. Report	Three of four routine cases pass; retain the failure.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An ordinary case is not automatically correct. Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires.

What this example does not establish

Keep the stated limit

Four routine tests do not cover equality, invalid **input**s or every relevant classroom condition.

Change one thing in the pictured case

Change one thing

Change: Prototype reports none for request 6 instead of P1.

Predicted result

Prediction: That run still fails.

Why the result follows

Reason: P2 can hold 6, so none is also wrong under the first-fit rule.

Ask while showing the picture

Ask while showing the picture · 8

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Test Normal Cases, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Four routine cases are chosen before testing. Three are correct.

2. State the decision to explain

Write the question to explain: What can be reported?

3. Reveal the worked reasoning

Reveal after **prediction**s: 3/4 on those cases. Preserve the failure and trace its cause. This does not guarantee unfamiliar-case performance.

Hold back the independent task

Keep for an independent attempt: Test a sorter on two clear pencil cards and two clear book cards. It gets one book wrong. Report the result and keep the failed case.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Q has 3 of 4 full passes and the fictional **baseline** 4 of 4. R4 selects B2 despite the quiet requirement. Dropping R4 answers how Q did on a selected three-case subset and conceals the failure in the original set.

Gaps include missing or invalid fields, equality at thresholds, accepted range endpoints and successful later-row selection under a varied catalogue.

Exit: this named version passed these four routine cases under the recorded conditions; all-**input** correctness and a speed advantage are not established.

Two distinct useful checks could be invalid-**input** handling and equal-time boundaries. Measuring speed would require a separate fair timing comparison.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve expected versus observed results and why disagreement is retained.
7 to 20 minutes	Read the contract and diagram; prepare W1 expectations and agree the full-pass criterion.
20 to 33 minutes	Work through the fictional Q table and W3, including the denominator and failed quiet condition.
33 to 45 minutes	Begin W2 partner runs with fixed versions and one trace per request.

Stage	Teacher action
Session 2 start to 8 minutes	Finish W2 baseline comparison and audit whether any cases or failures were omitted.
8 to 23 minutes	Complete the guided explanation and discuss coverage gaps without claiming universal reliability.
23 to 38 minutes	Collect W4 independently before offering support; require reasons and the denominator.
38 to 45 minutes	Complete W5 and the exit; retain versioned records for boundary testing next week.

Reduce help deliberately

Completion practice

Completion task: Operate your saved P1 cards and the manual **baseline** on all four R cases. Record **actual result**s and traces on your sheet.

Summarise full passes over all attempted cases for each method, name any failures and explain why a tie would not prove the prototype improves correctness. Do not invent an observed run.

Reasoning to listen for

Teacher check: A correctly executed P1 and **baseline** both produce B1, B2, no match, no match with correct reasons, giving 4 of 4 each.

Accept an honest record of different observed results when the pupil identifies and investigates the disagreements. Both methods must use the same requests, catalogue and **pass criterion**.

A tie establishes no correctness advantage on this set; an unperformed run must remain unperformed.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Testing routine cases. Proposed explanation: “A correct label is always a full pass”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: The agreed criterion also requires a correct explanation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why is none correct for 8 but incorrect for 6? Use both capacity cards.

Reasoned teacher answer

Neither capacity 3 nor 7 reaches 8. Capacity 7 does reach 6, so P2 is required for 6.

Changed-case reasoning: That run still fails. P2 can hold 6, so none is also wrong under the first-fit rule.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Test Normal Cases investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Run five planned tests and log expected versus actual.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the **expected result** separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR A failed test should be removed from the score.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A correct label is always a full pass: The agreed criterion also requires a correct explanation.

Remove a failed run because something went wrong: Keep and investigate it; missing required **output** cannot count as success.

Four passes prove all **input**s work: The conclusion is limited by the selected cases and conditions.

The prototype must beat the manual **baseline**: A correct simple **baseline** can tie or outperform a prototype; report the **evidence**.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Test Normal Cases questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

> **Teacher key · Worked case answer**
>
> 3/4 on those cases. Preserve the failure and trace its cause. This does not guarantee unfamiliar-case performance.

Independent question

> **Independent question · withhold the key**
>
> Test a sorter on two clear pencil cards and two clear book cards. It gets one book wrong. Report the result and keep the failed case.

Expected independent answer

> **Teacher key · Expected independent answer**
>
> It succeeds on 3 of 4 cards. Keep the mistaken book **input**, expected shelf **output** and actual **output**. Report the failure rather than dropping it from the total.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A failed test should be removed from the score.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Report 5/6, about 83.3%, on the six specified routine cases, with the failure documented. Investigate its cause and test a related new case after any revision.

If an exclusion is justified, disclose its reason and show results with and without it.

Explanations for the additional workbook examples

W1 Prepare the four cases

Workbook question · W1

Create the test record for R1 11 Yes, R2 9 No, R3 6 No and R4 9 Yes. Before running anything, write each expected **output**, explanation and purpose using catalogue B version 1.

Include blank places for observed **output**, observed reason, pass or fail and trace reference. State the version labels and **pass criterion**.

Teacher reasoning · W1

R1 B1 because time fits and it is quiet; early match. R2 B2 because B1 is too long, B2 fits and quiet is unrestricted; later match. R3 no match because every time exceeds 6; exhaustion.

R4 no match because B2 fails quiet while other rows exceed 9; both conditions. Record P1 version 1, B version 1 and the manual **baseline**. A full pass needs the required **output** and a correct explanation.

W2 Run a fair comparison

Workbook question · W2

Operate your saved P1 cards and the manual **baseline** on all four R cases. Record **actual result**s and traces on your sheet.

Summarise full passes over all attempted cases for each method, name any failures and explain why a tie would not prove the prototype improves correctness. Do not invent an observed run.

Teacher reasoning · W2

A correctly executed P1 and **baseline** both produce B1, B2, no match, no match with correct reasons, giving 4 of 4 each.

Accept an honest record of different observed results when the pupil identifies and investigates the disagreements. Both methods must use the same requests, catalogue and **pass criterion**.

A tie establishes no correctness advantage on this set; an unperformed run must remain unperformed.

W3 Keep the failed case in the total

Workbook question · W3

Use the fictional Q version 0 record. Calculate its full-pass fraction and the **baseline**'s fraction. Explain why R4 fails and why dropping it to report 3 of 3 misrepresents the original four-case set.

State what **evidence** you would inspect before changing Q.

Teacher reasoning · W3

Q passes 3 of 4; the **baseline** passes 4 of 4. R4 returns nonquiet B2 when quiet is required, so time alone is insufficient.

Removing R4 changes the tested set and hides its known failure; 3 of 3 is not the original set's result. Inspect the saved Q instructions and R4 trace to locate the missing quiet check, then label any repair separately.

W4 Audit a new test record

Workbook question · W4

Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10.

Select the first item with capacity at least the group size, or no match; invalid **input** needs clarification.

The saved prototype reports E1 for group 3 with reason "4 holds 3"; E1 for group 6 with reason "4 holds 6"; no match for group 9 with reason "both too small".

Write all **expected result**s, mark full passes, calculate the fraction and explain the failure. Give one untested **input** category. No booking occurs.

Teacher reasoning · W4

Expected group 3: E1 because 4 holds 3. Group 6: E2 because E1 is too small and 8 holds 6. Group 9: no match because both capacities are too small.

The first and third pass; the second fails both **output** and reason, giving 2 of 3 full passes. The record suggests a capacity-check problem but the saved instructions and trace are needed to locate it.

Missing, nonwhole or out-of-range **input** is untested, as is equality with a capacity.

W5 State what the evidence leaves open

Workbook question · W5

Explain why catalogue B cannot produce B3 or B4 as its first **fitting output** even if you try many ordinary requests. Distinguish checking a row during a failed search from successfully selecting it.

Write a limited conclusion for a P1 run that passed all four R cases and name a useful next test category.

Teacher reasoning · W5

B3 needs at least **12 minutes** and is quiet, so earlier B1 at 10 also fits whenever B3 fits. B4 needs 15 and is nonquiet, so a compatible request permits earlier B1 as well.

Both later rows may be checked in no-match runs but are never first fits in catalogue B. Report 4 of 4 full passes on set R with named versions; broader reliability is unestablished.

Next examine boundaries, invalid **input**s or an explicitly labelled catalogue variation.

Assess independent transfer

Independent question · withhold the key

Selected case: Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10.

Select the first item with capacity at least the group size, or no match; invalid **input** needs clarification.

The saved prototype reports E1 for group 3 with reason “4 holds 3”; E1 for group 6 with reason “4 holds 6”; no match for group 9 with reason “both too small”.

Write all **expected result**s, mark full passes, calculate the fraction and explain the failure. Give one untested **input** category. No booking occurs.

Selected case answer

> **Teacher key · Selected case answer**
>
> Expected group 3: E1 because 4 holds 3. Group 6: E2 because E1 is too small and 8 holds 6. Group 9: no match because both capacities are too small.
>
> The first and third pass; the second fails both **output** and reason, giving 2 of 3 full passes. The record suggests a capacity-check problem but the saved instructions and trace are needed to locate it.
>
> Missing, nonwhole or out-of-range **input** is untested, as is equality with a capacity.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Test Normal Cases to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Create and run the sorter's normal-case test plan with unique IDs and version label.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 34: Test Edge Cases and Fairness. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide blank column headings and a separate copy of the requirements. Ask pupils to check time and quiet separately before combining them.

Permit a spoken explanation recorded by the teacher, while retaining the pupil's independent first attempt. Extension: explain why repeated R1 runs do not add **coverage** of a missing condition.

W4 must independently produce E1, E2 and no match, identify the group-6 capacity failure and report 2 of 3 full passes without diagnosing unseen code.

After support, use a fresh record: groups 2, 7 and 10 produce observed E1 with “4 holds 2”, E2 with “8 holds 7”, and E2 with “8 holds 10”. Expected E1, E2, no match gives 2 of 3.

Require the pupil to identify the final failure and propose an untested validation case.

Week 34 tests equality boundaries, accepted ranges, invalid **input**s and explicitly labelled catalogue variations, then revisits what the **evidence** can support.

Plan the delayed return

Suggested return points in this level: ai-1/34, ai-1/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Test Normal Cases lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Plan routine cases with **expected result**s derived from the requirements

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Operate your saved P1 cards and the manual **baseline** on all four R cases. Record **actual result**s and traces on your sheet.

Summarise full passes over all attempted cases for each method, name any failures and explain why a tie would not prove the prototype improves correctness. Do not invent an observed run.

Supported practice answer

Reasoned practice answer: A correctly executed P1 and **baseline** both produce B1, B2, no match, no match with correct reasons, giving 4 of 4 each.

Accept an honest record of different observed results when the pupil identifies and investigates the disagreements. Both methods must use the same requests, catalogue and **pass criterion**.

A tie establishes no correctness advantage on this set; an unperformed run must remain unperformed.

Check every learner before discussion: Use the worked case for Testing routine cases. Proposed explanation: “A correct label is always a full pass”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: The agreed criterion also requires a correct explanation.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Prototype reports none for request 6 instead of P1. Require a **prediction** and a reason before comparing. **Expected result**: That run still fails. Reason: P2 can hold 6, so none is also wrong under the first-fit rule.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

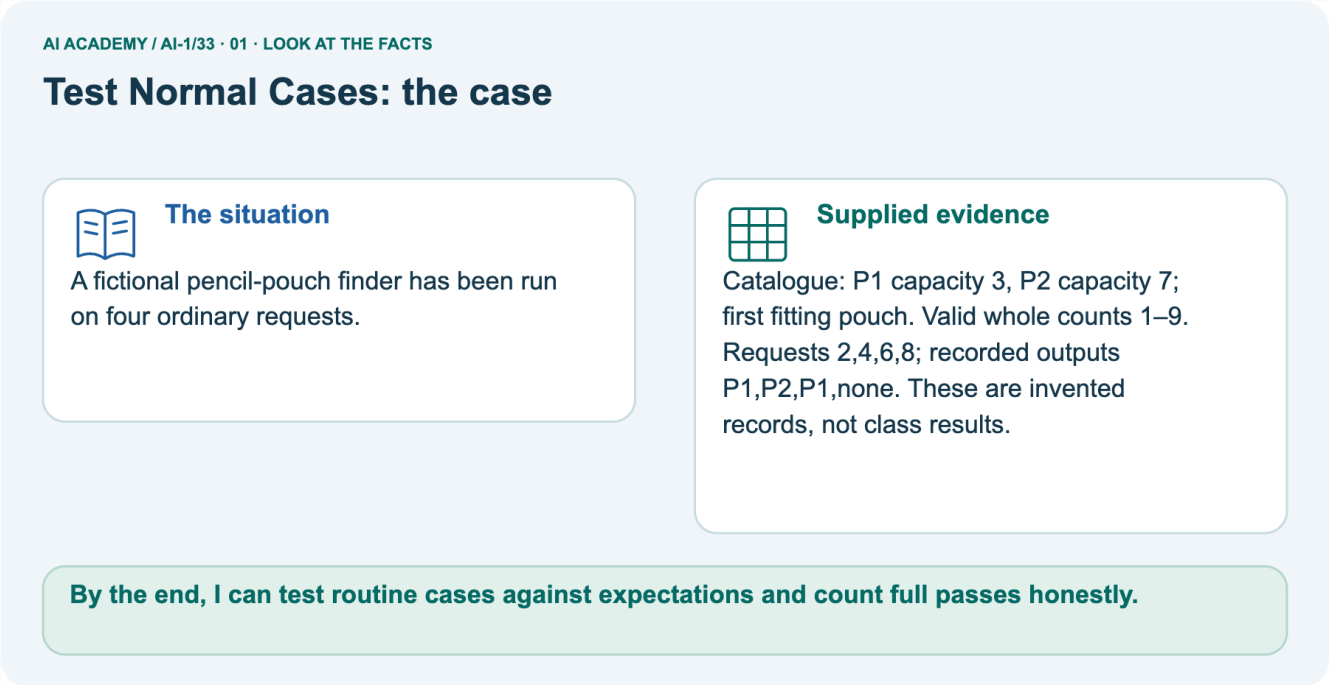
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Test Normal Cases graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Test Normal Cases: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional pencil-pouch finder has been run on four ordinary requests. Supplied facts: Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9. Requests 2,4,6,8; recorded **output**s P1,P2,P1,none.

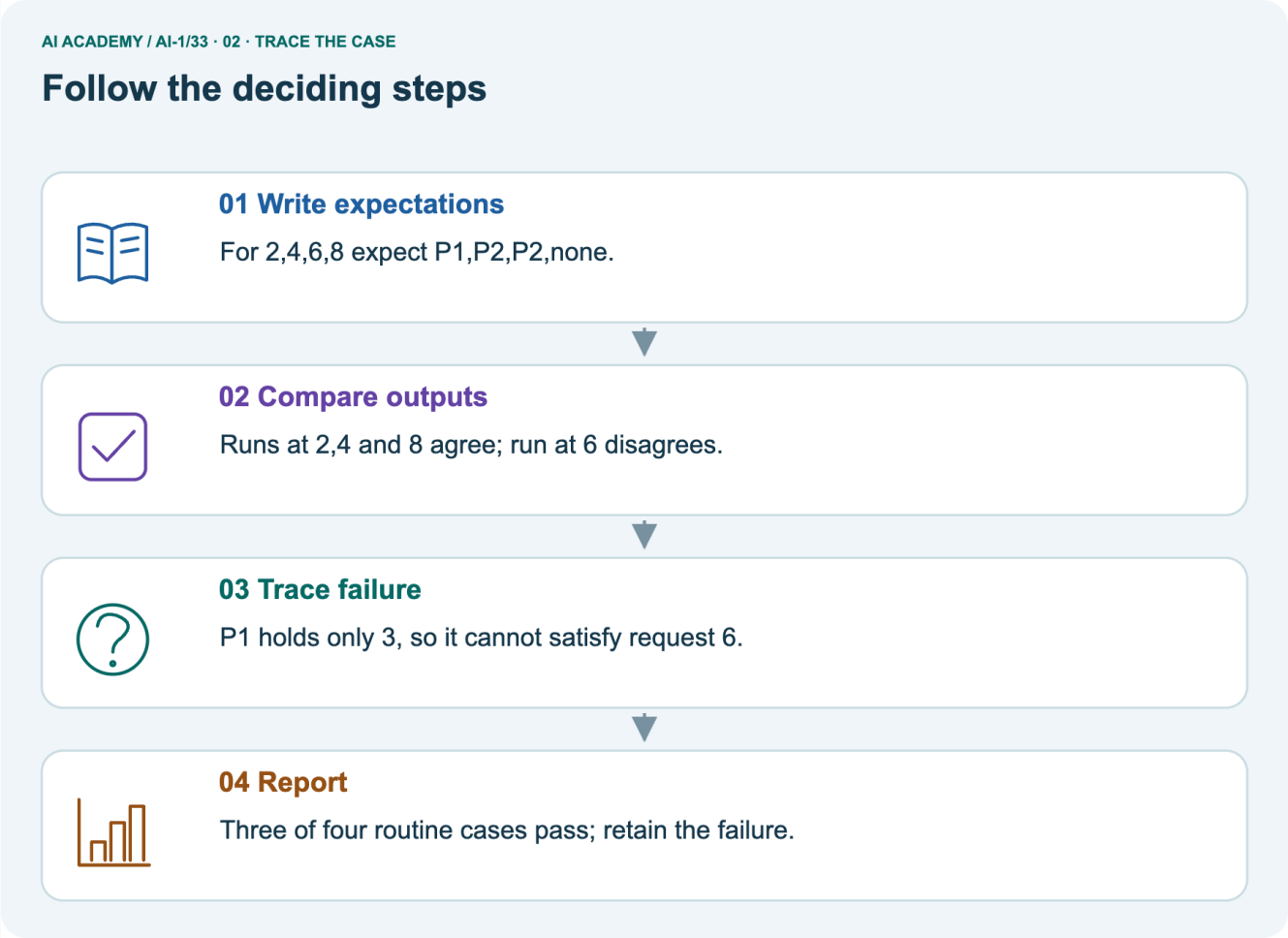
These are invented records, not class results.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

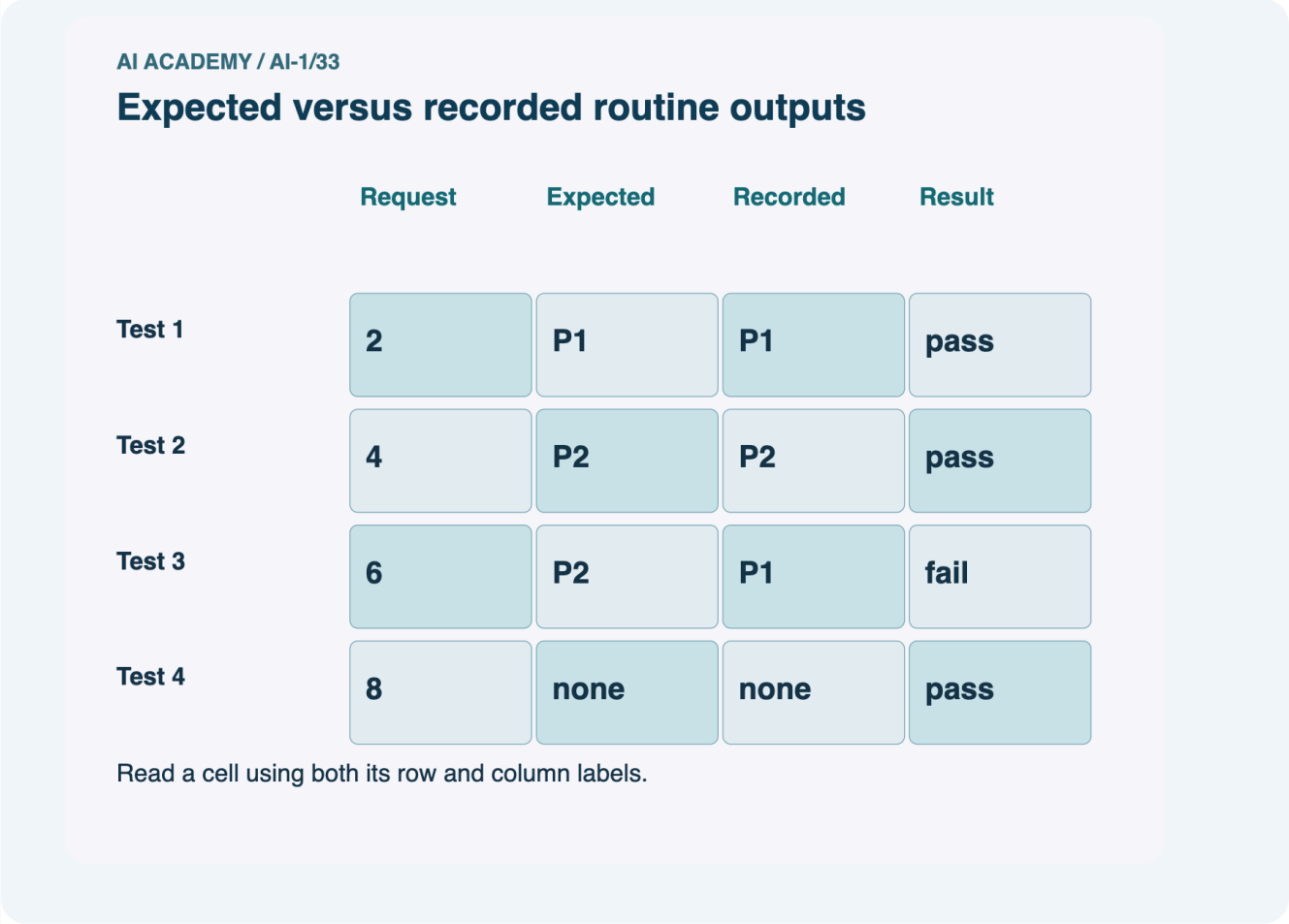
Write expectations: For 2,4,6,8 expect P1,P2,P2,none. → Compare **output**s: Runs at 2,4 and 8 agree; run at 6 disagrees. → Trace failure: P1 holds only 3, so it cannot satisfy request 6. → Report: Three of four routine cases pass; retain the failure.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Expected versus recorded routine outputs



Follow the labelled picture with the matching explanation. Open the vector diagram

Expected versus recorded routine **output**s. Test 1: Request = 2, Expected = P1, Recorded = P1, Result = pass. Test 2: Request = 4, Expected = P2, Recorded = P2, Result = pass.

Test 3: Request = 6, Expected = P2, Recorded = P1, Result = fail. Test 4: Request = 8, Expected = none, Recorded = none, Result = pass.

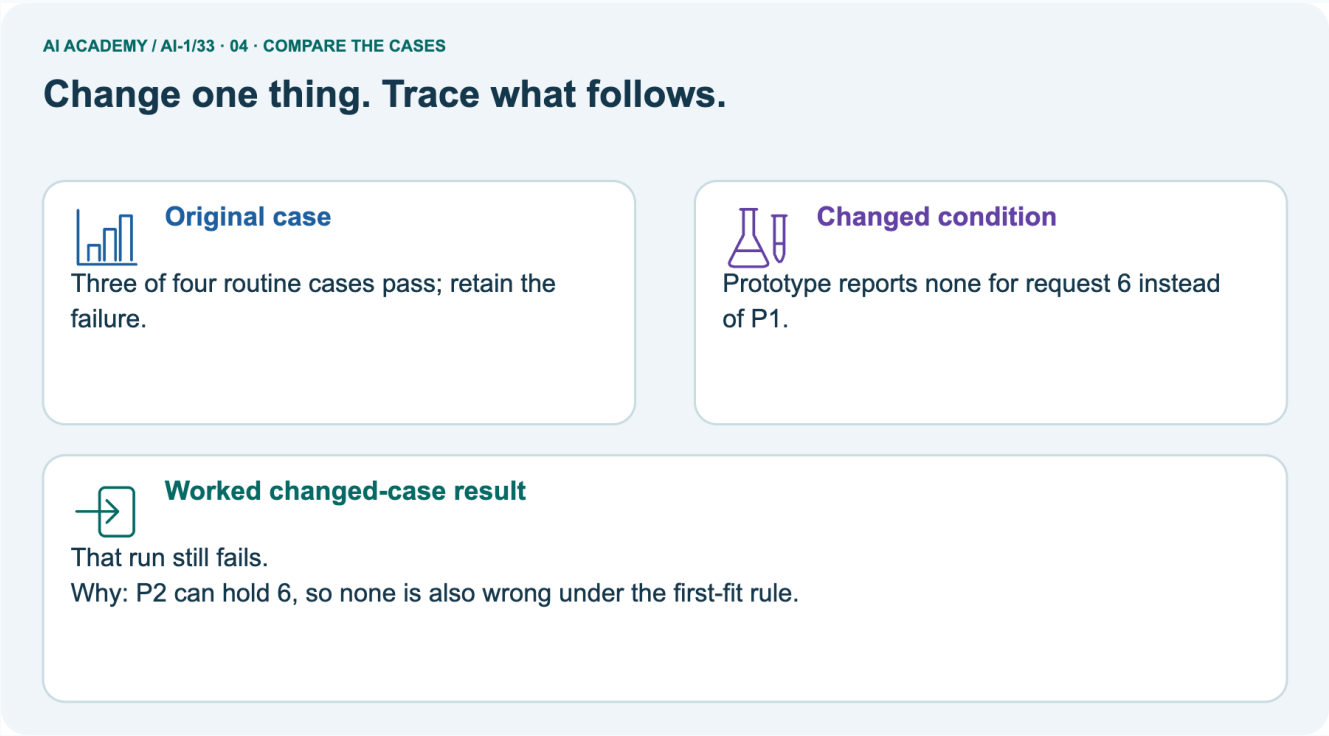
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three of four routine cases pass; retain the failure. Changed condition: Prototype reports none for request 6 instead of P1. **Expected result**: That run still fails.

Reason: P2 can hold 6, so none is also wrong under the first-fit rule.

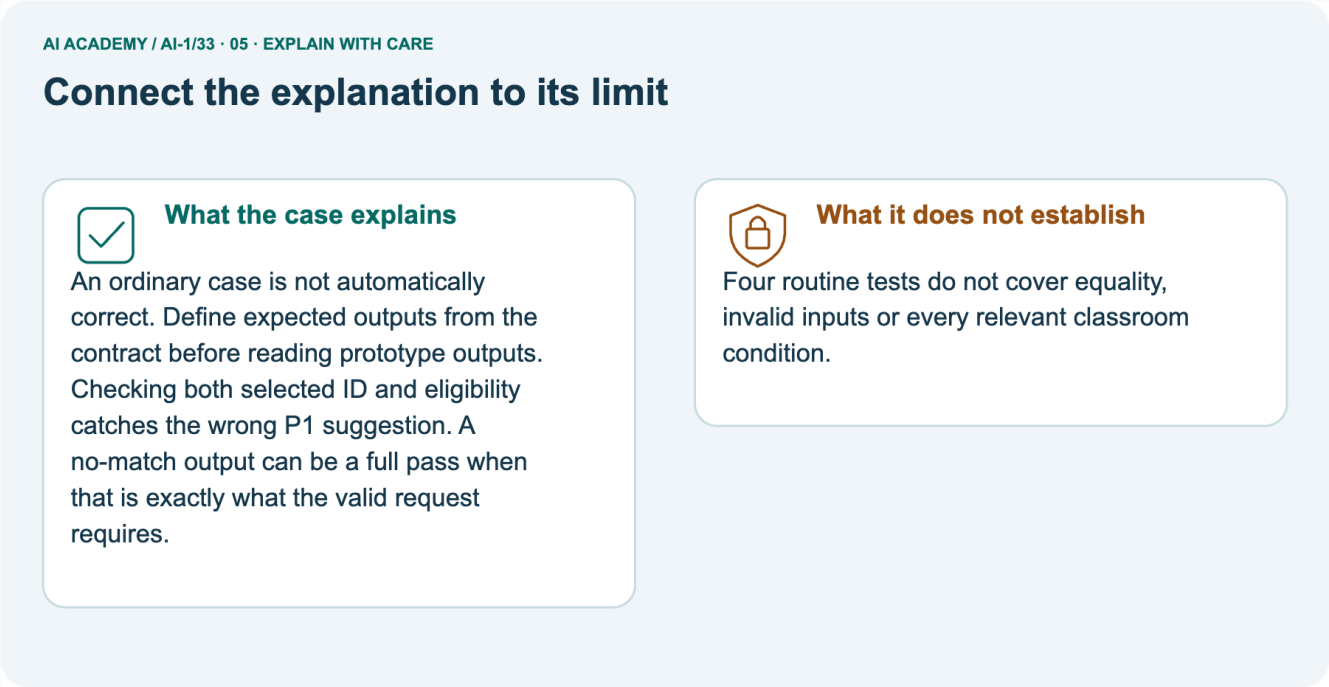
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: P2 can hold 6, so none is also wrong under the first-fit rule.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An ordinary case is not automatically correct. Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires. Limit: Four routine tests do not cover equality, invalid **input**s or every relevant classroom condition.

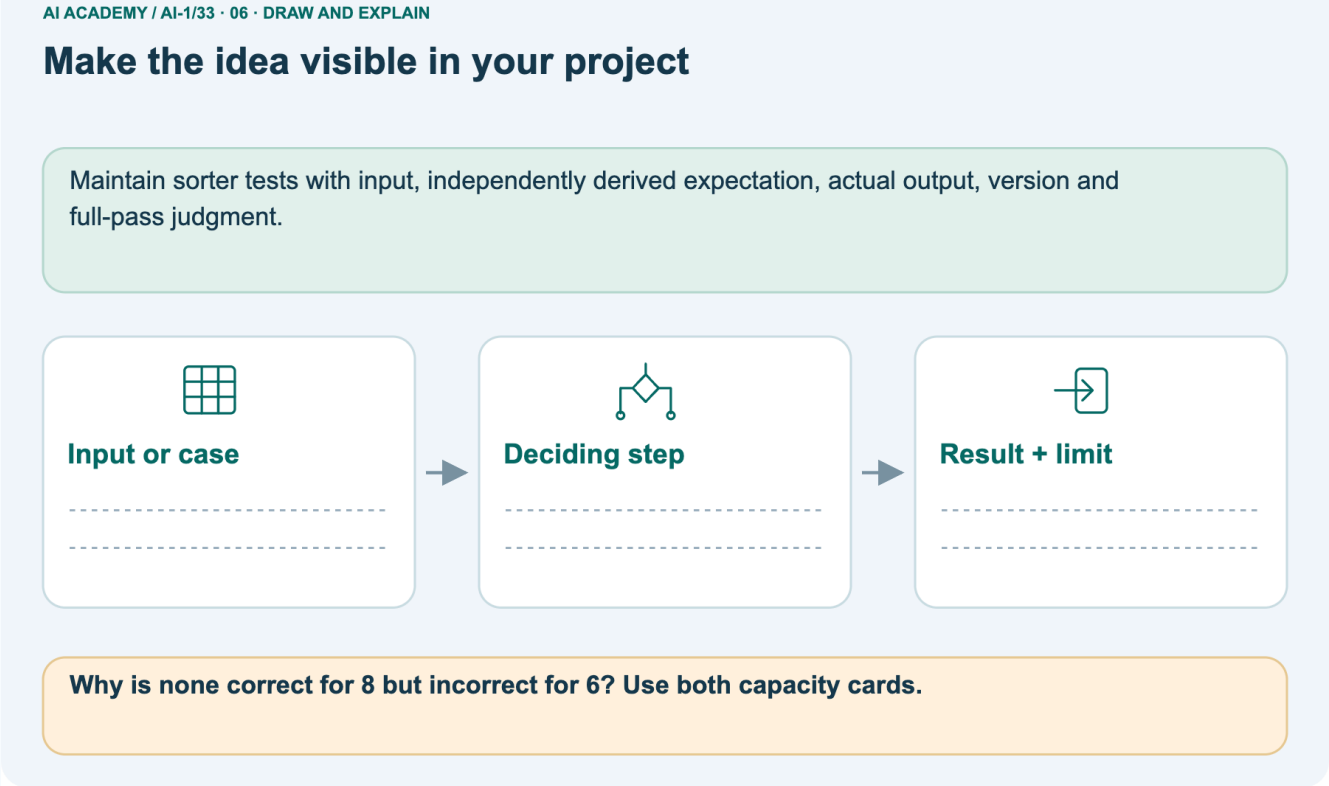
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Four routine tests do not cover equality, invalid **input**s or every relevant classroom condition.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment. Draw your **input**, deciding step, result and limit.

Supported question: Why is none correct for 8 but incorrect for 6? Use both capacity cards.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Neither capacity 3 nor 7 reaches 8. Capacity 7 does reach 6, so P2 is required for 6.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.